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Saveetha Institute of Medical And Technical Sciences
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Rule-Based Enterprise Workflow Optimization for Identifying and Reducing Task Bottlenecks Using Functions and Dynamic Memory Allocation

CSA0201 – C PROGRAMMING

Guided by,
Dr. Charlyn Pushpa Latha G

Presented by,
Ashira Bridget Susan S – 192524316
Priyavarsheni M V - 192524428

Introduction

- Modern enterprises handle hundreds of workflow tasks daily across multiple departments.
- Delays in task completion can reduce operational efficiency and productivity.
- Identifying workflow bottlenecks early helps organizations improve resource utilization.
- This project implements a rule-based workflow optimization framework using the c programming language.
- Functions and dynamic memory allocation are utilized to efficiently manage and analyze workflow data.



Literature Survey

Authors & Year	Title	Key Contribution	Relevance to Project	Authors & Year
Yu & Yao (2026)	<i>Agents and Workflows</i>	Explores intelligent agents and workflow management using large language models for task execution and automation.	Provides the conceptual foundation for agent-based workflow optimization and intelligent task management.	Yu & Yao (2026)
Shoba et al. (2026)	<i>Hierarchical ML Ensemble Engine for Dynamic Project Management Optimization and Resource Intelligence</i>	Proposes an ML ensemble framework for optimizing project management and resource allocation dynamically.	Demonstrates techniques for improving resource utilization and workflow efficiency in enterprise environments.	Shoba et al. (2026)
Kosińska et al. (2025)	<i>Rule-based Execution of Management Actions in Cloud-native Environments</i>	Introduces a rule-based approach for automating management actions in cloud-native systems.	Supports the use of constraint-based and rule-driven mechanisms for workflow execution and bottleneck reduction.	Kosińska et al. (2026)
Yandamuri (2024)	<i>AI-Enabled Workflow Automation and Predictive Analytics for Enterprise Operations Management</i>	Presents AI-driven workflow automation integrated with predictive analytics for enterprise operations.	Highlights the role of AI in predicting performance trends and enhancing operational decision-making.	Yandamuri (2026)

Problem Statement

- Enterprise workflows often experience delays due to inefficient task execution.
- Manual monitoring of large task datasets is time-consuming and error-prone.
- Bottlenecks are difficult to identify without systematic analysis.
- Organizations require a lightweight and explainable solution to detect workflow inefficiencies.
- There is a need for a rule-driven system that can identify, classify, and reduce task bottlenecks.



Objectives

- ❖ **Analyze workflow tasks and identify delayed activities.**
- ❖ **Detect bottlenecks using predefined business rules.**
- ❖ **Classify bottlenecks into minor, moderate, and severe categories.**
- ❖ **Recommend suitable workflow management strategies.**
- ❖ **Simulate optimization and measure performance improvement.**
- ❖ **Generate analytical reports for workflow evaluation.**



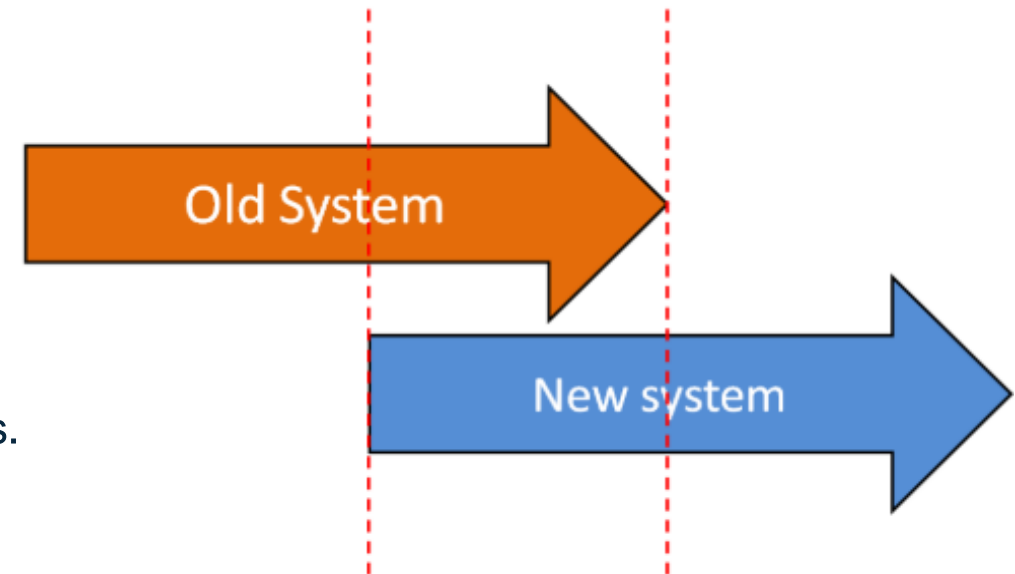


Existing System

- Workflow monitoring is largely manual.
- Delays are identified only after operational issues occur.
- Limited visibility into department-wise performance.
- Lack of structured bottleneck classification mechanisms.
- Decision-making depends heavily on human observation.
- Difficult to scale for large datasets.

Proposed System

- ✓ A rule-based enterprise workflow optimization framework.
- ✓ Automatically analyzes workflow datasets.
- ✓ Detects bottlenecks based on task delays.
- ✓ Assigns workflow management methods using business rules.
- ✓ Simulates optimization outcomes to estimate performance gains.
- ✓ Generates department-wise and task-wise analytical reports.
- ✓ Uses dynamic memory allocation for scalable data handling.



Module 1 – Intelligent Workflow Analysis Engine

FUNCTIONS PERFORMED

- ❖ Loads workflow dataset from CSV files.
- ❖ Stores records dynamically using memory allocation.
- ❖ Calculates time differences between estimated and actual completion times.
- ❖ Detects bottlenecks automatically.
- ❖ Categorizes delays into:
 - Minor delay
 - Moderate delay
 - Severe delay
- ❖ Determines workflow efficiency status.

OUTCOME

Provides accurate identification of workflow bottlenecks and operational inefficiencies.

```
===== RULE-BASED WORKFLOW OPTIMIZER =====  
1. Load dataset  
2. Run rule engine (detect + reduce bottlenecks)  
3. View workflow summary  
4. View department breakdown  
5. View top bottlenecks  
6. Validate rule engine against dataset  
7. Export full report to CSV  
8. Exit  
=====
```

Module 2 – Optimization & Decision Support Engine

FUNCTIONS PERFORMED

- ❖ Evaluates task priority levels.
- ❖ Assigns workflow management methods:
 - Optimized
 - Hybrid
 - Manual
- ❖ Simulates post-optimization task execution.
- ❖ Calculates expected performance improvement.
- ❖ Generates workflow summaries and departmental reports.
- ❖ Validates rule-engine accuracy against the dataset.

OUTCOME

Supports decision-making and recommends strategies to reduce workflow delays.

===== RULE-BASED WORKFLOW OPTIMIZER =====

1. Load dataset
2. Run rule engine (detect + reduce bottlenecks)
3. View workflow summary
4. View department breakdown
5. View top bottlenecks
6. Validate rule engine against dataset
7. Export full report to CSV
8. Exit

=====

Advantages & Limitations

Advantages

- Simple and easy to understand.
- Fully explainable rule-based decisions.
- No machine learning dependency.
- Fast execution with low computational overhead.
- Dynamic memory improves scalability.
- Generates accurate workflow reports.
- Achieved 100% validation accuracy on the dataset.

Limitations

- Performance depends on predefined rules.
- Rules must be updated when business processes change.
- Cannot automatically learn from new patterns.
- Limited adaptability compared to advanced AI systems.

Future Scope

- Integration with real-time workflow monitoring systems.
- Development of graphical dashboards for visualization.
- Hybrid integration with machine learning models.
- Cloud-based workflow analytics platform.
- Automated resource allocation recommendations.
- Real-time bottleneck alerts and notifications.
- Support for larger enterprise datasets and distributed systems.



Conclusion

- A rule-based enterprise workflow optimization framework was successfully developed.
- The system effectively identifies and categorizes workflow bottlenecks.
- Dynamic memory allocation enables efficient management of workflow records.
- Optimization rules help improve task execution strategies.
- Department-wise and workflow-wide insights support better decision-making.
- The project demonstrates how rule-based intelligence can enhance enterprise workflow efficiency in a transparent and cost-effective manner.



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